Counting

Permutations and combinations

Definition	Expression
Order all n distinct objects	$N = n(n-1)\cdots(2)(1) = n!; \quad 0! = 1$
Order k; no replacement	$P_{n,k} = {}_nP_k = n(n-1)\cdots(n-k+1) = \frac{n!}{(n-k)!}; \quad 0 \le k \le n$
Choose k; unordered, no repeats	$\binom{n}{k} = {}_nC_k = C(n,k) = \frac{P(n,k)}{k!} = \frac{n!}{k!(n-k)!}; \quad 0 \le k \le n$
Order r from n; repeats	n^r
Order multiset; n_i identical/type	$\frac{n!}{n_1!n_2!\cdots n_k!}; \quad n_1+\cdots+n_k = n, \quad n_i \in \mathbb{Z}_{\ge 0}$
Unordered r of n; unlimited repeats	$\binom{n+r-1}{r}; \quad n, r \in \mathbb{Z},\ n \ge 1,\ r \ge 0$

Binomial theorem $(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}; \quad n \in \mathbb{Z}_{\ge 0}$

Counting the hands $N = \binom{52}{4} = 2{,}598{,}960$

Hand / pattern	Choices
High card $\{x,y,z,u,v\}$	$\left(\binom{13}{5} - 10\right)\left(4^5 - 4\right) = 1{,}302{,}540$
Pair $\{x,x,y,z,u\}$	$13\binom{4}{2}\binom{12}{3}4^3 = 1{,}098{,}240$
Two pairs $\{x,x,y,y,z\}$	$\binom{13}{2}\binom{4}{2}^2 \cdot 11 \cdot 4 = 123{,}552$
Three of a kind $\{x,x,x,y,z\}$	$13\binom{4}{3}\binom{12}{2}4^2 = 54{,}912$
Straight $\{x,x+1,\ldots,x+4\}$	$10(4^5 - 4) = 10{,}200$
Flush $\{x,y,z,u,v\}$	$4\left(\binom{13}{5} - 10\right) = 5{,}108$
Full House $\{x,x,x,y,y\}$	$13 \cdot 12\binom{4}{3}\binom{4}{2} = 3{,}744$
Four of a kind $\{x,x,x,x,y\}$	$13\binom{4}{4} \cdot 12 \cdot 4 = 624$
Straight Flush $\{x,x+1,\ldots,x+4\}$	$9 \cdot 4 = 36$
Royal Flush $\{10,J,Q,K,A\}$	$1 \cdot 4 = 4$

Rules of Probability

De-Morgan's Law

Complement: $A^c = \bar{A}$.

$$\overline{A \cup B} = \bar{A} \cap \bar{B}$$
$$\overline{A \cap B} = \bar{A} \cup \bar{B}$$

Generalized DeMorgan's laws

$$\left(\bigcup_{i\in I} A_i\right)^c = \bigcap_{i\in I} A_i^c$$
$$\left(\bigcap_{i\in I} A_i\right)^c = \bigcup_{i\in I} A_i^c$$

I: any index set.

Rule (Inclusion-exclusion)

$$P\left(\bigcup_{i=1}^{n} A_i\right) = \sum_i P(A_i) - \sum_{i<j} P(A_i \cap A_j) + \sum_{i<j<k} P(A_i \cap A_j \cap A_k) - \cdots + (-1)^{n+1} P(A_1 \cap \cdots \cap A_n)$$

Count each unordered intersection once.

Pairwise mutually exclusive: $A_i \cap A_j = \varnothing$ for $i \neq j$:

$$P\left(\bigcup_{i=1}^{n} A_i\right) = \sum_{i=1}^{n} P(A_i)$$

Probability: Part III

Independence: Definition

$$P(A \cap B) = P(A)P(B)$$

Equivalent: $P(A \mid B) = P(A)$ if $P(B) > 0$; exchange A, B if $P(A) > 0$.

Disjoint $A \cap B = \varnothing$ with $P(A), P(B) > 0 \Rightarrow$ dependent.

Notes

Mutual: every subcollection of distinct indices, $2 \le k \le n$; pairwise alone is insufficient.

$$P\left(\bigcap_{j=1}^{k} A_{i_j}\right) = \prod_{j=1}^{k} P(A_{i_j})$$

Conditional independence $(P(C) > 0)$:

$$P(A \cap B \mid C) = P(A \mid C)P(B \mid C)$$

Does not imply unconditional independence.

The multiplicative rule of probability

$$P[A \cap B] = P[A]P[B \mid A], \quad P[A] > 0,$$
$$= P[B]P[A \mid B], \quad P[B] > 0.$$
$$P(A_1 \cap \cdots \cap A_n) = P(A_1)P(A_2 \mid A_1) \cdot P(A_3 \mid A_1 \cap A_2) \cdots \cdot P(A_n \mid A_1 \cap \cdots \cap A_{n-1})$$

Condition on **all earlier events**; each conditioning event has positive probability.

Rule: Law of Total Probability

Finite/countable partition: exclusive = no overlap; exhaustive = covers S.

$$A_i \cap A_j = \varnothing \ (i \neq j), \quad \bigcup_i A_i = S, \quad \sum_i P(A_i) = 1$$
$$P(B) = \sum_i P(A_i)P(B \mid A_i)$$

Use $P(A_i) > 0$; omit zero-probability cases. Independence not required.

Bayes Rule / General Bayes Theorem

Same partition; $P(B) > 0$.

$$P(A_i \mid B) = \frac{P(B \mid A_i)P(A_i)}{P(B)} = \frac{P(B \mid A_i)P(A_i)}{\sum_j P(B \mid A_j)P(A_j)}$$

Term / notation	Meaning
Prior $P(A_i)$	Case probability before B.
Likelihood $P(B \mid A_i)$	Evidence probability given case.
Evidence $P(B)$	Overall observation probability.
Posterior $P(A_i \mid B)$	Case probability after B.

Parameter: which A_i is true. **Data:** observed B.

Posterior $\propto$ **likelihood** $\times$ **prior.**

Sensitivity and specificity

A = disease, B = positive test; $P(A), P(\bar{A}) > 0$.

$$\text{Sensitivity} = P(B \mid A), \quad \text{Specificity} = P(\bar{B} \mid \bar{A})$$

False-positive rate: $P(B \mid \bar{A}) = 1 - \text{specificity}$.

Random Variables

Definition

Random variable: outcome $\to$ real number; $X : S \to \mathbb{R}, \quad \omega \mapsto X(\omega)$.

Probability mass function (PMF)

Discrete: finite/countable support; point probabilities.

$$p(x) = P(X = x) = P\{\omega \in S : X(\omega) = x\}$$
$$0 \le p(x) \le 1, \qquad \sum_x p(x) = 1$$
$$P(X \in E) = \sum_{x \in E} p(x)$$

Cumulative Distribution Function (CDF)

$$F_X(x) = P(X \le x), \qquad 0 \le F_X(x) \le 1$$
$$\lim_{x \to -\infty} F(x) = 0, \qquad \lim_{x \to \infty} F(x) = 1$$
$$a \le b \Rightarrow F(a) \le F(b), \qquad \lim_{t \to x^+} F(t) = F(x)$$

All types: **nondecreasing, right-continuous**; fully specifies distribution.

Probability Density Function (PDF)

$$f(x) \ge 0, \qquad \int_{-\infty}^{\infty} f(x)\,dx = 1$$
$$P(a < X < b) = \int_a^b f(x)\,dx = F(b) - F(a)$$
$$F(x) = \int_{-\infty}^{x} f(t)\,dt, \qquad f(x) = F'(x)$$

Derivative: where f continuous (a.e. generally).

$$P(x < X \le x + h) \approx f(x)h$$

Small $h > 0$, f continuous at x. Density $\neq$ point probability; may exceed 1 or decrease.

$P(X = a) = 0$; endpoints/isolated density values do not affect probability. Density $\Rightarrow$ continuous CDF; converse need not hold.

Mixed Random Variables

Masses + density: atoms $p(u) = P(X = u) > 0$; $f \ge 0$.

$$P(X \in E) = \sum_{u \in E} p(u) + \int_E f(x)\,dx$$
$$\sum_u p(u) + \int_{-\infty}^{\infty} f(x)\,dx = 1$$
$$F(x) = \sum_{u \le x} p(u) + \int_{-\infty}^{x} f(t)\,dt$$

CDF derivative $\to$ density part; jumps $\to$ atoms.

The distribution function F(x)

$$F(a-) = \lim_{u \to a^-} F(u) = P(X < a)$$
$$P(X = a) = F(a) - F(a-)$$

No atom: $F(a-) = F(a)$; always $F(a+) = F(a)$.

Event $(a < b)$	Probability
$a < X \le b$	$F(b) - F(a)$
$a \le X \le b$	$F(b) - F(a-)$
$a < X < b$	$F(b-) - F(a)$
$a \le X < b$	$F(b-) - F(a-)$
$X > a$	$1 - F(a)$
$X \ge a$	$1 - F(a-)$

For Discrete Random Variables

$$F(x) = \sum_{u \le x} p(u)$$

CDF jumps = masses; derivatives do not recover them.

Jointly Distributed Random Variables

Joint probability function

$$p(x,y) = P(X = x, Y = y)$$
$$0 \le p(x,y) \le 1, \qquad \sum_x \sum_y p(x,y) = 1$$
$$P[(X,Y) \in A] = \sum_{(x,y) \in A} p(x,y)$$

Support: possible pairs; impossible pairs have probability 0.

Definition: Marginal Distributions

$$p_X(x) = P(X = x) = \sum_y p(x,y)$$
$$p_Y(y) = P(Y = y) = \sum_x p(x,y)$$
$$0 \le p_X(x), p_Y(y) \le 1, \quad \sum_x p_X(x) = \sum_y p_Y(y) = 1$$

Conditional Distributions

$$p_{X|Y}(x \mid y) = P(X = x \mid Y = y) = \frac{p(x,y)}{p_Y(y)}$$
$$p_{Y|X}(y \mid x) = P(Y = y \mid X = x) = \frac{p(x,y)}{p_X(x)}$$

Require $p_Y(y) > 0$, $p_X(x) > 0$, respectively.

$$0 \le p_{X|Y}(x \mid y) \le 1, \quad \sum_x p_{X|Y}(x \mid y) = 1$$
$$0 \le p_{Y|X}(y \mid x) \le 1, \quad \sum_y p_{Y|X}(y \mid x) = 1$$
$$P(X \in E \mid Y = y) = \sum_{x \in E} p_{X|Y}(x \mid y)$$

Fixed condition: PMF totals 1; whole conditional table need not.

Joint and marginal from conditionals

$$p(x,y) = p_Y(y)p_{X|Y}(x \mid y) = p_X(x)p_{Y|X}(y \mid x)$$
$$p_X(x) = \sum_y p_{X|Y}(x \mid y)p_Y(y)$$

Positive-probability conditioning values only.

Relative Risk

Group	Event	No event
Exposed	a	b
Unexposed	c	d

$$\text{RR} = \frac{\text{Risk in Exposed Group}}{\text{Risk in Unexposed Group}} = \frac{a/(a+b)}{c/(c+d)}$$

Finite ratio: $a + b > 0$, $c + d > 0$, $c > 0$. Exposed risk:

$RR = 1$: equal; $RR > 1$: higher; $RR < 1$: lower.

Bivariate Continuous Distributions

Definition: Joint probability density

$$f(x,y) \ge 0, \qquad \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\,dx\,dy = 1$$
$$P[(X,Y) \in A] = \iint_A f(x,y)\,dx\,dy$$

Event ∩ support; $f = 0$ outside support. Density may exceed 1; $P(X = a, Y = y) = 0$.

Multiple Integration

Volume under $z = f(x,y)$. Inner variable varies; outer variable fixed. Reverse order: **same region, new bounds**.

Rectangle: $a \le x \le b,\ c \le y \le d$

$$\int_c^d \left[\int_a^b f(x,y)\,dx\right] dy = \int_a^b \left[\int_c^d f(x,y)\,dy\right] dx$$

Integration over nonrectangular regions

Type I: $a \le x \le b, \quad g_1(x) \le y \le g_2(x)$

$$\iint_A f(x,y)\,dx\,dy = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x,y)\,dy\,dx$$

Type II: $c \le y \le d, \quad a(y) \le x \le b(y)$

$$\iint_A f(x,y)\,dx\,dy = \int_c^d \int_{a(y)}^{b(y)} f(x,y)\,dx\,dy$$

Inner bounds may depend on outer variable. Split where boundaries change.

Definition: Marginal densities

$$f_X(x) = \int_{-\infty}^{\infty} f(x,y)\,dy, \quad f_Y(y) = \int_{-\infty}^{\infty} f(x,y)\,dx$$

Each: nonnegative, integral 1, zero outside support. Bounds: **joint support**.

Definition of Conditional Density

$$f_{Y|X}(y \mid x) = \frac{f(x,y)}{f_X(x)}, \qquad f_X(x) > 0$$
$$f_{X|Y}(x \mid y) = \frac{f(x,y)}{f_Y(y)}, \qquad f_Y(y) > 0$$

Density ratio $(P(X = x) = 0)$; each fixed admissible x:

$$f_{Y|X}(y \mid x) \ge 0, \quad \int_{-\infty}^{\infty} f_{Y|X}(y \mid x)\,dy = 1$$
$$P(a < Y < b \mid X = x) = \int_a^b f_{Y|X}(y \mid x)\,dy$$

Use conditional support; swap X, Y for reverse condition.

Definition - Mutual Independence

$$f(x,y) = f_X(x)f_Y(y) \quad \text{(almost everywhere)}$$

Discrete: $f \to p$. Conditioning leaves distribution unchanged when conditioning density/mass is positive.

Expectation (Mean) of a Random Variable

Expectation (Mean) of a Random Variable

Mean $\mu = E(X)$: probability-weighted average.

$$E(X) = \sum_x xp(x) = \sum_i x_i p(x_i)$$
$$E(X) = \int_{-\infty}^{\infty} xf(x)\,dx$$

Full support; finite if $E(|X|) < \infty$. Need not be a possible value.

The Uniform distribution

$a < b$.

$$f(x) = \begin{cases} \dfrac{1}{b-a}, & a \le x \le b, \\ 0, & \text{otherwise.} \end{cases}$$
$$E(X) = \frac{a+b}{2}$$

$X \sim U(0,1)$: $f(x) = 1$ on $[0,1]$, 0 elsewhere.

$$\mu_k = \frac{1}{k+1}$$
$$\mu_1 = \frac{1}{2}, \quad \mu_2 = \frac{1}{3}, \quad \mu_3 = \frac{1}{4}$$
$$\text{var}(X) = \frac{1}{12}, \quad \sigma = \frac{1}{\sqrt{12}}$$
$$m(t) = \begin{cases} \dfrac{e^t - 1}{t}, & t \neq 0, \\ 1, & t = 0. \end{cases}$$

MGF: all real t.

Expectation of Functions of Random Variables

$$E[g(X)] = \begin{cases} \sum_x g(x)p(x), & \text{discrete,} \\ \int_{-\infty}^{\infty} g(x)f(x)\,dx, & \text{continuous.} \end{cases}$$

Finite if $E[|g(X)|] < \infty$; generally $E[g(X)] \neq g(E(X))$.

Moments of Random Variables

Raw moments

$$\mu_k = E(X^k) = \begin{cases} \sum_x x^k p(x), & \text{discrete,} \\ \int_{-\infty}^{\infty} x^k f(x)\,dx, & \text{continuous.} \end{cases}$$

Integer $k \ge 1$; finite if $E(|X|^k) < \infty$.

$$\mu = \mu_1 = E(X), \quad \mu_2 = E(X^2), \quad \mu_3 = E(X^3)$$

Definition: Central moments

$$\mu_k^0 = E[(X - \mu)^k], \qquad \mu_1^0 = 0$$
$$\text{var}(X) = \mu_2^0 = E[(X - \mu)^2] = \sigma^2$$
$$\sigma^2 = \begin{cases} \sum_x (x - \mu)^2 p(x), & \text{discrete,} \\ \int_{-\infty}^{\infty} (x - \mu)^2 f(x)\,dx, & \text{continuous.} \end{cases}$$
$$\sigma = \sqrt{\text{var}(X)}, \qquad z = \frac{x - \mu}{\sigma} \ (\sigma > 0)$$

$\text{var}(X) \ge 0$; zero iff $P(X = \mu) = 1$. Variance: squared units; SD: original units.

Third central moment: $\mu_3^0 = E[(X - \mu)^3]$.

Properties of Expected Value

a, b, c constant; required moments finite.

$$E[c] = c, \qquad E[aX + b] = aE[X] + b$$
$$E[g(X) + h(X)] = E[g(X)] + E[h(X)]$$

$E[aX^2 + b] = aE(X^2) + b$.

$$\text{var}(X) = E(X^2) - [E(X)]^2 = \mu_2 - \mu_1^2$$
$$\text{var}(aX + b) = a^2 \,\text{var}(X)$$
$$\text{SD}(aX + b) = |a|\sigma$$

Using marginals

$$E(X^k) = \sum_x x^k p_X(x) \quad \text{or} \quad \int x^k f_X(x)\,dx$$

Likewise for Y; independence not required.

Moment generating functions

$M_X = m_X$.

$$m_X(t) = E[e^{tX}] = \begin{cases} \sum_x e^{tx} p(x), & \text{discrete,} \\ \int_{-\infty}^{\infty} e^{tx} f(x)\,dx, & \text{continuous.} \end{cases}$$

Exists: finite on $(-\delta, \delta)$ for some $\delta > 0$. Bounded support: all real t; otherwise check domain.

Finite MGF $\Rightarrow$ all absolute moments finite; **converse false**. Any infinite moment rules it out.

$$m_X(0) = 1, \quad m_X(t) = 1 + \mu_1 t + \frac{\mu_2}{2!}t^2 + \frac{\mu_3}{3!}t^3 + \cdots$$
$$m_X^{(k)}(0) = \frac{d^k}{dt^k} m_X(t)\Big|_{t=0} = \mu_k = E(X^k)$$

Differentiate, then set $t = 0$: **raw** moments.

$$E(X) = m_X'(0), \quad E(X^2) = m_X''(0), \quad E(X^3) = m_X'''(0)$$
$$\text{var}(X) = m_X''(0) - [m_X'(0)]^2$$

Coefficient of $t^k \times k!$ = μ_k; finite MGF sum: coefficient of e^{tx} = $P(X = x)$.

$$X_1, \ldots, X_n \text{ independent}, \ Y = \sum_i X_i \ \Rightarrow \ m_Y(t) = \prod_i m_{X_i}(t)$$

On a common finite-MGF interval.

Special Distributions

Bernoulli Distribution

One trial: $X = 1$ success, $X = 0$ failure; $q = 1 - p$.

$$X \sim \text{Bernoulli}(p), \quad p(x) = \begin{cases} q, & x = 0, \\ p, & x = 1. \end{cases}$$
$$E(X) = p, \quad \text{Var}(X) = pq, \quad m_X(t) = q + pe^t$$

$0 \le p \le 1$. Throughout: PMFs = 0 outside stated support.

Summary

Binomial distribution

X: successes in n independent trials, same p.

$$X \sim \text{Binom}(n,p), \quad p(x) = \binom{n}{x} p^x q^{n-x}$$

$x = 0, 1, \ldots, n$; n nonnegative integer; $0 \le p \le 1$.

$$P(X \ge r) = 1 - \sum_{x=0}^{r-1} \binom{n}{x} p^x q^{n-x}$$
$$m_X(t) = (q + pe^t)^n \quad (t \in \mathbb{R})$$
$$E(X) = np, \quad E(X^2) = n^2p^2 + npq, \quad \text{Var}(X) = npq$$
$$M_X'(t) = n(pe^t + q)^{n-1}pe^t \implies E(X) = np$$

$M_X = m_X$; derivative above: $n \ge 1$. $n = 0$: $M_X(t) = 1$, $M_X'(t) = 0$.

Differentiation & Integration

Essential derivatives

a, b, c: constants; u, v: functions. $f'(x) = \frac{df}{dx}$: slope; f'', f''': second, third derivatives.

Function	Derivative
c	0
x^n	$\frac{d}{dx}x^n = nx^{n-1}$
x	1
x^2, x^3	$2x$, $3x^2$
$1/x = x^{-1}$	$-1/x^2$
$\sqrt{x} = x^{1/2}$	$1/(2\sqrt{x})$ for $x > 0$
e^x	e^x
e^{ax+b}	ae^{ax+b}
$\ln\lvert x\rvert$	$1/x$ for $x \neq 0$

$$\frac{d}{dx}[au(x) + bv(x)] = au'(x) + bv'(x)$$

Constant n; powers where the function and derivative exist.

Chain rule

$$\frac{d}{dx}g(u(x)) = g'(u(x))u'(x)$$

Exponential: $(e^{u(x)})' = e^{u(x)}u'(x)$.

Product and quotient rules

Rule	Derivative
Product	$(uv)' = u'v + uv'$
Quotient	$\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}, \ v \neq 0$

Antiderivatives / fundamental theorem

$G' = f$; C: arbitrary constant.

$$\int f(x)\,dx = G(x) + C$$

Continuous f:

$$\int_a^b f(x)\,dx = [G(x)]_a^b = G(b) - G(a)$$
$$\frac{d}{dx}\int_a^x f(t)\,dt = f(x).$$

Piecewise: split where formula changes.

Essential integrals

Integrand	Antiderivative
c	$cx + C$
x^r	$\frac{x^{r+1}}{r+1} + C, \ r \neq -1$
x, x^2	$x^2/2 + C$, $x^3/3 + C$
$1/x$	$\ln\lvert x\rvert + C, \ x \neq 0$
e^x	$e^x + C$
e^{ax+b}	$\frac{e^{ax+b}}{a} + C, \ a \neq 0$

$a = 0$: $e^{ax+b} = e^b$; integral $e^b x + C$.

Constants and sums under an integral

$U' = u$, $V' = v$:

$$\int [au(x) + bv(x)]\,dx = aU(x) + bV(x) + C.$$

Linearity also holds for definite integrals.

$$\int_a^b c\,dx = c(b - a)$$